

Class 1 Reserved-Slot Strategies: Day-Level Formulation

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1 Introduction

1.1 Purpose and Reader Map

This note is a companion to `first_two_class_simulation_note_v2.tex`. It keeps the same day-level notation, the same patient behavior objects, and the same metric interpretation, but it adds two booking strategies in which Class 1 has protected daily capacity.

Core Policy Question

The pooled FCFS baseline asks which patient arrives first. The reserved-slot strategies ask whether some daily capacity should be protected for Class 1 before the two classes compete for the remaining slots.

The three strategies compared in this note are:

- **Pooled FCFS:** every class competes for the same S daily slots.
- **Strict Class 1 reservation:** $Q = 10$ slots per day are reserved for Class 1 only; unused reserved slots stay empty.
- **Released Class 1 reservation:** Class 1 receives first access to the $Q = 10$ reserved slots; after all Class 1 arrivals have had that first pass, Class 2 may fill unused reserved slots.

1.1.1 What Changes from the FCFS Note

Object	FCFS baseline	Reserved-slot replacement
Daily capacity	one pooled capacity S	protected capacity Q plus general capacity $S - Q$
Calendar state	$X_{i,r}^D, N_r^D$	same $X_{i,r}^D$, plus audit-layer pool counts $X_{i,r,\text{res}}^D$ and $X_{i,r,\text{gen}}^D$
Offer rule	earliest residual day with $N_r^D < S$	earliest residual day with eligible pool capacity
No-offer logic	no pooled slot is available	no eligible pool is available for that class and policy
Service, cancellation, no-show	unchanged	unchanged

2 Model Formulation

2.1 Notation Inherited from v2

The simulation clock is still the day D . The basic objects are unchanged:

- S : total appointment slots available per day.
- H : rolling booking horizon in days.
- $\mathcal{I} = \{1, 2\}$: patient classes.
- $r \in \{0, 1, \dots, H - 1\}$: residual delay, where $r = 0$ means appointments scheduled for today.
- τ : original offered delay accepted by a booked patient.
- $X_{i,r}^D$: number of class- i patients scheduled for day $D + r$.
- $N_r^D = \sum_{i \in \mathcal{I}} X_{i,r}^D$: total scheduled patients on residual day r .

The same residual-delay/original-delay distinction remains:

Residual Delay vs. Original Delay

Residual delay r is part of the rolling calendar state and decreases when the day advances. Original delay τ is stored with the booking because balking and no-show behavior use the delay the patient was originally offered.

To make the policy formulas match the current implementation and remain compatible with the v2 notation, define the bookable residual-day set

$$\mathcal{R} \subseteq \{0, 1, \dots, H - 1\}.$$

In the current Python engine, same-day booking is allowed, so

$$\mathcal{R} = \{0, 1, \dots, H - 1\}.$$

If a future-only convention is desired, the same formulas apply after replacing $\mathcal{R}$ with $\{1, \dots, H - 1\}$.

2.2 Reservation Parameters

Let the protected class be

$$i^* = 1,$$

and let

$$Q = 10$$

be the number of Class 1 reserved slots per day. Since the baseline capacity is $S = 32$, the general pool has

$$S - Q = 22$$

slots per day.

The total day-level state $X_{i,r}^D$ is still the main state, but the reserved policies require an audit-layer pool label. Write

$$X_{i,r,\text{res}}^D = \text{number of class-}i \text{ patients scheduled in reserved slots on day } D + r,$$

and

$$X_{i,r,\text{gen}}^D = \text{number of class-}i \text{ patients scheduled in general slots on day } D + r.$$

Then

$$X_{i,r}^D = X_{i,r,\text{res}}^D + X_{i,r,\text{gen}}^D.$$

Define the pool occupancies

$$R_r^D = \sum_{i \in \mathcal{I}} X_{i,r,\text{res}}^D, \quad G_r^D = \sum_{i \in \mathcal{I}} X_{i,r,\text{gen}}^D,$$

so that

$$N_r^D = R_r^D + G_r^D.$$

The capacity constraints are

$$R_r^D \leq Q, \quad G_r^D \leq S - Q, \quad N_r^D \leq S.$$

Daily Capacity Split for the Implemented Strategy

Pool	Capacity	Eligibility
Reserved pool	$Q = 10$	Class 1 always; Class 2 only under the released-reservation policy after the Class 1 first pass
General pool	$S - Q = 22$	Class 1 overflow and Class 2
Total day	$S = 32$	all scheduled patients, regardless of pool label

2.3 Patient Behavior is Unchanged

The class behavior object remains

$$C_i = \{\lambda_i, b_i(\tau), \phi_i, \xi_i(\tau)\}.$$

The reservation policies do not change arrival rates, balking, cancellation, or no-show behavior. They only change which residual day and which capacity pool an arriving patient is offered.

What the Reservation Strategy Does Not Change

For a class- i patient offered delay τ , the balking probability remains $b_i(\tau)$. For a booked class- i patient scheduled in the future, the daily cancellation probability remains ϕ_i . At service time, the no-show probability remains $\xi_i(\tau)$.

3 Booking Rules

3.1 Pooled FCFS Baseline

The pooled FCFS baseline searches the residual days in order and offers the first day with open total capacity:

$$\tau_{\text{FCFS}} = \min \{r \in \mathcal{R} : N_r^D < S\}.$$

If the set is empty, the patient receives no offer.

If the patient accepts, the simulator increments

$$X_{i, \tau_{\text{FCFS}}}^D \leftarrow X_{i, \tau_{\text{FCFS}}}^D + 1$$

and stores the audit record

$$(i, \tau_{\text{FCFS}}).$$

FCFS in One Line

Under pooled FCFS, class labels do not affect the offer rule. They affect behavior after the offer through $b_i(\tau)$, ϕ_i , and $\xi_i(\tau)$, but not the capacity search itself.

3.2 Strict Class 1 Reservation

Under strict reservation, Class 1 has first access to the reserved pool on each day. If the reserved pool is already full on the earliest eligible day, Class 1 can compete for general slots. Class 2 can use only the general pool. Unused reserved slots stay empty and are carried forward as protected future capacity until the appointment day arrives.

3.2.1 Class 1 Offer Rule

For a Class 1 arrival, the strict policy searches residual days in increasing r . On each residual day, it checks the reserved pool first and the general pool second:

$$\tau_{1, \text{strict}} = \min \{r \in \mathcal{R} : R_r^D < Q \text{ or } G_r^D < S - Q\}.$$

The pool label is

$$z_{1,\text{strict}}(r) = \begin{cases} \text{res}, & R_r^D < Q, \\ \text{gen}, & R_r^D = Q \text{ and } G_r^D < S - Q. \end{cases}$$

If both pools are unavailable for every $r \in \mathcal{R}$, the Class 1 patient receives no offer. If the patient accepts, the corresponding pool count is incremented:

$$X_{1,\tau,\text{res}}^D \leftarrow X_{1,\tau,\text{res}}^D + 1 \quad \text{or} \quad X_{1,\tau,\text{gen}}^D \leftarrow X_{1,\tau,\text{gen}}^D + 1.$$

3.2.2 Class 2 Offer Rule

For a Class 2 arrival under strict reservation, the reserved pool is invisible:

$$\tau_{2,\text{strict}} = \min \{r \in \mathcal{R} : G_r^D < S - Q\}.$$

If the set is empty, the Class 2 patient receives no offer even if some reserved capacity remains unused:

$$R_r^D < Q \quad \text{does not help Class 2 under strict reservation.}$$

If the patient accepts, the simulator increments

$$X_{2,\tau,\text{gen}}^D \leftarrow X_{2,\tau,\text{gen}}^D + 1.$$

Strict Reservation Interpretation

Strict reservation protects Class 1 access at the cost of possible empty protected capacity. It is not a utilization-maximizing policy; it is an access-protection policy.

3.3 Released Class 1 Reservation

The released policy keeps the same $Q = 10$ protected slots, but it changes the within-day order of the booking step. Class 1 gets the first pass on the reserved pool. After that first pass, remaining patients compete for open capacity, and Class 2 may backfill protected slots that Class 1 did not reserve.

3.3.1 First Pass: Class 1 Reserved Slots

Let the randomized arrival sequence on day D be

$$A^D = (a_1^D, a_2^D, \dots, a_{M^D}^D),$$

where each $a_k^D \in \mathcal{I}$. In the first pass, the policy processes only entries with

$$a_k^D = 1.$$

Each Class 1 patient is offered the earliest residual day with reserved capacity:

$$\tau_{1,\text{first}} = \min \{r \in \mathcal{R} : R_r^D < Q\}.$$

If that set is nonempty and the patient accepts, the simulator increments

$$X_{1,\tau,\text{res}}^D \leftarrow X_{1,\tau,\text{res}}^D + 1.$$

If no reserved capacity is available, that Class 1 arrival is not counted as no-offer yet. It moves to the second pass and competes for general capacity.

3.3.2 Second Pass: General Capacity and Released Reserved Capacity

After the Class 1 first pass, the second pass processes:

- all Class 2 arrivals;
- Class 1 arrivals that could not reserve a protected slot in the first pass.

Class 1 overflow uses only the general pool:

$$\tau_{1,\text{release,gen}} = \min \{r \in \mathcal{R} : G_r^D < S - Q\}.$$

Class 2 first searches the general pool and then may use released reserved capacity:

$$\tau_{2,\text{release}} = \min \{r \in \mathcal{R} : G_r^D < S - Q \text{ or } R_r^D < Q\}.$$

The pool label for Class 2 is

$$z_{2,\text{release}}(r) = \begin{cases} \text{gen}, & G_r^D < S - Q, \\ \text{res}, & G_r^D = S - Q \text{ and } R_r^D < Q. \end{cases}$$

Released Reservation Interpretation

Released reservation is a compromise. Class 1 receives priority access to Q protected daily slots, but unused protected capacity can still be converted into Class 2 access after the Class 1 first pass.

4 Visual Calendar Logic

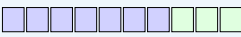
4.1 One Residual Day

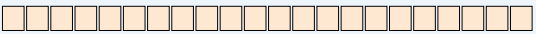
The reservation logic is easiest to see on a single residual day r . The day has $Q = 10$ protected slots and $S - Q = 22$ general slots.

Pool Occupancy on One Residual Day

Suppose $S = 32$, $Q = 10$, and the post-cancellation state for one residual day is

$$R_r^D = 7, \quad G_r^D = 22, \quad N_r^D = 29.$$

reserved $Q = 10$  $R_r^D = 7$, three protected slots open

general $S - Q = 22$  $G_r^D = 22$, no general slots open

Under strict reservation, a Class 1 patient can be offered this day, but a Class 2 patient cannot. Under released reservation after the Class 1 first pass, a Class 2 patient can use one of the three open protected slots.

4.2 Residual-Day Search

Comparing the Earliest Feasible Day

Take $H = 4$, $S = 32$, $Q = 10$, and the post-cancellation state below.

Count	day D $r = 0$	day $D + 1$ $r = 1$	day $D + 2$ $r = 2$	day $D + 3$ $r = 3$
R_r^D	10	8	10	6
G_r^D	22	22	20	13
N_r^D	32	30	30	19

The implied offers are:

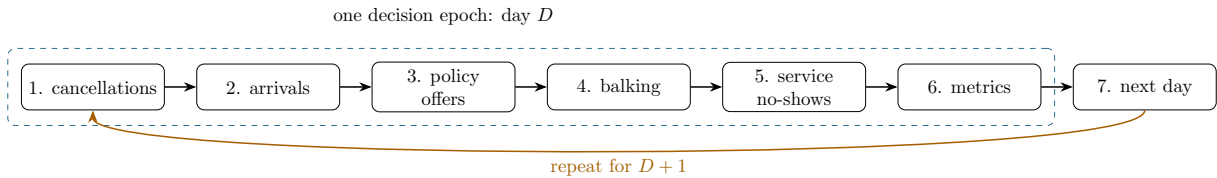
Policy and class	Earliest offer	Reason
Pooled FCFS, either class	$r = 1$	$N_1^D = 30 < 32$
Strict reservation, Class 1	$r = 1$, reserved	$R_1^D = 8 < 10$
Strict reservation, Class 2	$r = 2$, general	$G_1^D = 22$ is full, but $G_2^D = 20 < 22$
Released reservation, Class 2 after first pass	$r = 1$, reserved if unused	released protected slots can be backfilled

5 Daily Simulation Mechanics

5.1 Unchanged Daily Order

The daily event order remains the same as in the v2 note:

1. **Compute cancellations for the day.** Future appointments independently cancel with probability ϕ_i .
2. **Simulate daily arrivals.** $M_i^D \sim \text{Poisson}(\lambda_i)$.
3. **Generate a randomized within-day arrival order.**
4. **Apply the configured offer rule.** This is the only step changed by the reserved-slot strategies.
5. **Simulate balking.** A class- i patient offered delay τ accepts with probability $1 - b_i(\tau)$.
6. **Simulate no-shows and service for day D .**
7. **Compute metrics and advance to day $D + 1$.**



5.2 Strict Reservation Day

Under strict reservation, the randomized arrival order is processed once. The difference from pooled FCFS is that the feasible capacity set depends on the patient's class:

$$\mathcal{F}_{1,\text{strict}}^D = \{(r, \text{res}) : r \in \mathcal{R}, R_r^D < Q\} \cup \{(r, \text{gen}) : r \in \mathcal{R}, G_r^D < S - Q\},$$

where reserved capacity is checked before general capacity on the same residual day. For Class 2,

$$\mathcal{F}_{2,\text{strict}}^D = \{(r, \text{gen}) : r \in \mathcal{R}, G_r^D < S - Q\}.$$

If $\mathcal{F}_{i,\text{strict}}^D$ is empty, that patient receives no offer. The simulator then continues to later arrivals because a later Class 1 arrival may still have protected capacity even when a Class 2 arrival has no general capacity.

5.3 Released Reservation Day

Under released reservation, the day has two booking passes.

1. **Class 1 first pass.** Class 1 arrivals try to reserve protected slots:

$$\mathcal{F}_{1,\text{first}}^D = \{(r, \text{res}) : r \in \mathcal{R}, R_r^D < Q\}.$$

If this set is empty, the patient moves to the second pass instead of being counted as no-offer.

2. **Backfill and overflow pass.** Class 1 overflow competes for general capacity. Class 2 competes for general capacity first and then released protected capacity:

$$\mathcal{F}_{2,\text{release}}^D = \{(r, \text{gen}) : r \in \mathcal{R}, G_r^D < S - Q\} \cup \{(r, \text{res}) : r \in \mathcal{R}, R_r^D < Q\}.$$

This two-pass rule is the mathematical version of “Class 1 fills reserved slots first; if they are not filled, Class 2 may fill them later.”

6 Comparison to Pooled FCFS

6.1 Policy Effects

Feature	Pooled FCFS	Strict Class 1 reservation	Released Class 1 reservation
Capacity structure	all S slots pooled	Q protected, $S - Q$ general	Q protected first, then releasable
Class 1 access	competes only by arrival order	protected first, then general overflow	protected first, then general overflow
Class 2 access	can use any open slot	can use only general slots	can use general slots and unused protected slots after first pass
Unused reserved slots	not applicable	may remain empty	can be backfilled by Class 2
Utilization tendency	highest when demand is sufficient and no-shows are low	can fall if Class 1 demand is not enough to use Q	closer to pooled FCFS because unused protection can be recovered
Equity/access tendency	neutral in the offer rule; differences come from behavior	shifts access toward Class 1	shifts access toward Class 1, but less rigidly

6.2 Expected Metric Patterns

The main metrics are unchanged:

$$\rho^{\text{attended}} = \frac{\text{served slots}}{S \times \text{measured days}}, \quad \text{served rate}_i = \frac{\text{served}_i}{\text{arrivals}_i}.$$

The reservation strategies mainly affect the path to those metrics:

$$\text{offer rule} \rightarrow \tau \rightarrow b_i(\tau) \rightarrow \text{booked patients} \rightarrow \phi_i, \xi_i(\tau) \rightarrow \text{served patients}.$$

Interpretation of the Comparison

Strict reservation can improve Class 1 access while lowering aggregate utilization if protected slots are not filled. Released reservation tests whether most of the Class 1 access protection can be kept while reducing the utilization loss by letting Class 2 backfill unused protected capacity.

6.3 No-Offer Interpretation

The no-offer event has a slightly different meaning across the strategies:

Policy	No-offer meaning
Pooled FCFS	no day in $\mathcal{R}$ satisfies $N_r^D < S$
Strict reservation, Class 1	no day has either $R_r^D < Q$ or $G_r^D < S - Q$
Strict reservation, Class 2	no day has $G_r^D < S - Q$, even if $R_r^D < Q$ somewhere
Released reservation, Class 1	no protected slot is available in the first pass and no general slot is available in the second pass
Released reservation, Class 2	no general slot and no released protected slot are available in the second pass

7 Implementation Mapping

7.1 Config Objects

The new implementation fields map directly to the notation:

Code field	Notation	Meaning
slots_per_day	S	total daily capacity
reserved_class_id	i^*	protected class; here $i^* = 1$
reserved_slots_per_day	Q	protected daily capacity; here $Q = 10$
release_reserved_slots	release flag	false for strict reservation, true for released reservation
Booking.reserved_slot	pool label z	true means the appointment was placed in the reserved pool

7.2 Code Path

The policy logic appears in the booking step:

- `find_earliest_open_day` computes the earliest feasible (r, z) pair.
- `process_daily_arrivals` applies either the one-pass rule or the released two-pass rule.
- `reserved_only=True` implements the Class 1 first pass.
- `general_only=True` implements Class 1 overflow into the general pool.
- `release_reserved=True` allows Class 2 to use unused protected slots after the first pass.

Review Against the Implemented Reserved-Slot Model

- The original day-level state notation $X_{i,r}^D$, N_r^D , r , τ , S , and H is unchanged.
- The new audit-layer pool counts R_r^D and G_r^D split the same daily capacity into reserved and general pools.
- $Q = 10$ protected slots are reserved for Class 1 out of $S = 32$.
- In strict reservation, unused protected slots stay unavailable to Class 2.
- In released reservation, Class 1 gets the protected first pass and Class 2 can backfill unused protected capacity afterward.
- Cancellation, no-show, service, and metric definitions remain the same as in the FCFS v2

note.

8 Conclusion

The reservation strategies are not separate patient behavior models. They are changes to the offer rule inside the same day-level appointment simulator. Pooled FCFS uses one capacity constraint

$$N_r^D < S.$$

Strict Class 1 reservation replaces that with class-dependent eligibility over

$$R_r^D < Q \quad \text{and} \quad G_r^D < S - Q.$$

Released reservation keeps the protected first pass for Class 1, then relaxes the strict rule by letting Class 2 use protected capacity that Class 1 did not reserve.

The comparison should therefore be read as an access/utilization tradeoff:

$$\text{more Class 1 protection} \quad \Longleftrightarrow \quad \text{less automatic pooling of daily capacity.}$$